

Intergenerational Housing and Fertility: A Two-Page Model Summary

Textbook version of Part I

Purpose. The model describes how housing market institutions change completed fertility. The primitive is simple: children use consumption goods and housing space. If family-sized space is hard for young households to reach, children become expensive even when posted housing prices do not change.

Housing environment. There is one consumption good and one aggregate stock $\bar{H}$ of divisible floorspace. A household can occupy this floorspace as a renter, R , or as an owner, O . Tenure matters because the two modes offer different size menus:

$$\mathcal{H}^R = \{0 < h \leq \bar{h}^R\}, \quad \mathcal{H}^O = \{0 < h \leq \bar{h}^O\}, \quad \bar{h}^R < \bar{h}^O.$$

The baseline follows the one-price floorspace discipline: a unit of space has flow user cost q in either tenure, and the asset price is $P = q/\rho_p$, with $\rho_p = r + \delta^O + \tau^p$. Tenure changes the menu and financing constraints, not the baseline per-unit cost of space. Supply extensions with tenure-specific costs can be added later; then q is replaced by the relevant tenure user cost.

Young households. A young household i has income y_i , liquid assets a_i , and inheritance exposure B_i . With after-tax bequest share $x = 1 - \tau^b$, pledgeable resources are $A_i(x) = a_i + xB_i + T_i^b(x)$. These resources back a down payment; they are not additional flow income.

If the household has n_i children and occupies h_i , adult residual consumption and housing are $c_i = C_i - \chi_i n_i$ and $s_i = h_i - \kappa n_i$. The child requirements are χ_i units of consumption and κ units of space. Preferences are $u_i = \log c_i + \alpha \log s_i + \beta \log n_i$. In either baseline tenure branch, the budget can be written as

$$c_i + q s_i + (\chi_i + \kappa q) n_i = y_i.$$

This identity is the core economics: the full private cost of one more child is the child consumption requirement plus the market value of the space the child occupies.

The renter problem is this branch problem with $h_i \leq \bar{h}^R$. The owner problem adds financing constraints:

$$(1 - \phi) P h_i \leq A_i(x), \quad q h_i \leq \psi y_i, \quad h_i \leq \bar{h}^O.$$

Here ϕ is the financed share and ψ is the payment-to-income limit. The constraints collapse into one effective owner cap:

$$H_i^O = \min \left\{ \bar{h}^O, \frac{A_i(x) \rho_p}{(1 - \phi) q}, \frac{\psi y_i}{q} \right\}, \quad H_i^R = \bar{h}^R.$$

The household solves both branch problems and chooses $W_i^Y = \max\{W_i^R, W_i^O\}$. Entry is logit:

$$\pi_i^E = \frac{\exp(W_i^Y / \kappa_E)}{\exp(W_i^Y / \kappa_E) + \exp(\bar{W}^E / \kappa_E)},$$

where $\bar{W}^E$ is the outside value.

Old owners. Old households are incumbent owners. Old owner j has income m_j , initial housing H_j^0 , utility $\log c_j^O + \gamma \log h_j^O + \mathcal{B}_j(e_j)$, and a possible incumbent property-tax discount $L_j^\tau = \tau^p P - \tilde{\tau}_j \tilde{P}_j \geq 0$. The old budget is

$$c_j^O + e_j + (q - L_j^\tau) h_j^O = m_j + q H_j^0, \quad 0 \leq h_j^O \leq H_j^0.$$

Economically, L_j^τ is a subsidy to staying put. It is not old utility and not a real resource saving.

Competitive equilibrium. Given q , P , and policies, young households choose entry, tenure, housing, consumption, and fertility; old owners choose how much owner housing to retain. The single floorspace market clears:

$$\bar{M} \int \pi_i^E [\mathbf{1}\{o_i = R\} h_i^R + \mathbf{1}\{o_i = O\} h_i^O] dG_Y(i) + \int h_j^O dF_O(j) = \bar{H}.$$

This is aggregate clearing, not pairwise assignment. The market can clear even if constrained young families value an additional unit of space more than old incumbents who keep it.

Marginal values and fertility. Let $\zeta_i^m \geq 0$ be the goods-equivalent value of relaxing the binding family-housing cap in branch m . It is zero when the household is unconstrained. The marginal values are

$$\frac{\alpha c_i^R}{s_i^R} = q + \zeta_i^R, \quad \frac{\alpha c_i^O}{s_i^O} = q + \zeta_i^O, \quad \frac{\gamma c_j^O}{h_j^O} = q - L_j^\tau.$$

Here ζ_i^O may come from the down payment, the payment-to-income limit, or the owner size menu. A constrained young household values family space above the market price, while a protected old incumbent can retain space she values below the market price.

The fertility condition is the most compact statement of the mechanism:

$$\frac{\beta c_i^m}{n_i^m} = \chi_i + \kappa(q + \zeta_i^m).$$

The child is priced at the household's own value of space. If the cap binds, $\zeta_i^m > 0$, and the housing part of child cost rises one-for-one with the value of constrained space.

Planner and efficiency. The planner has the same real stock, preferences, and size menus. It counts true old housing and bequest utility, not the tax discount L_j^τ . It can relax down-payment or payment-to-income constraints only through feasible instruments, with real cost included in Ξ^Π . With no direct social birth payoff, the planner's interior rules are

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = Q, \quad \frac{u_{h,j}^O}{u_{c,j}^O} = Q, \quad \frac{\beta c_i}{n_i} = \chi_i + \kappa Q.$$

At an interior competitive equilibrium, the planner scarcity value is $Q = q$. Hence the competitive allocation fails the planner's marginal conditions when young owner finance constraints are relaxable implementation constraints with $\zeta_i^{O,F} > 0$, or when old incumbents face $L_j^\tau > 0$. A marginal unit shifted from a protected old owner to a constrained young family has surplus

$$\zeta_i^{O,F} + L_j^\tau$$

before instrument costs. Real size menus are different: if the planner faces the same menu, a binding menu cap is real scarcity, not by itself an inefficiency.

Policy and fertility effects. For a fixed branch, let H be the effective family-housing cap. If H binds, increasing H raises fertility whenever the extra space effect dominates the consumption cost of buying more housing. A clean sufficient and sharp global condition is

$$\chi_i \leq \frac{\kappa q}{\alpha}.$$

This says that the child input bundle is not more consumption-intensive than the adult residual bundle. Under exact equal scaling, every relaxation of a strictly binding cap raises fertility until the cap stops binding.

For a small policy z , the fixed-price, fixed-tenure local decomposition is

$$\left. \frac{dN}{dz} \right|_{q,o,A} = \bar{M} \sum_m \int_{\mathcal{B}_m} \underbrace{\pi_i^E}_{\text{entered mass}} \underbrace{\left[\Psi_i^m + (n_i^m - \bar{n}_i^E) \frac{1 - \pi_i^E}{\kappa_E} \Theta_i^m \right]}_{\text{fertility response plus entry response}} \underbrace{P_i^m(z)}_{\text{pass-through to } H_i^m} dG_Y(i).$$

Here $\mathcal{B}_m$ is the set of capped types in active branch m , $P_i^m = \partial H_i^m / \partial z$, $\Psi_i^m = \partial n_i^m / \partial H_i^m$, and $\Theta_i^m = \partial W_i^m / \partial H_i^m = \lambda_i^m \zeta_i^m$. The formula says: exposed constrained mass, times pass-through to the binding cap, times the fertility response, plus the entry response. Policies that do not relax the binding component of H_i^m have no effect through this channel.

Tenure switching adds a boundary term because tenure is discrete. In the baseline, both tenures price space at q , so an indifferent household has the same allocation on both sides of $W_i^R = W_i^O$; the switching jump is zero and the fixed-tenure formula is complete. If tenure wedges create $q^O \neq q^R$, the sign is theoretical: owner-access switching reinforces fertility when ownership is the expensive large-space mode ($q^O > q^R$) and reverses when ownership is the cheaper small-space mode ($q^O < q^R$). The boundary mass is quantitative.